13 Write $x^2 + 6x - 7$ in the form $(x + a)^2 + b$ where a and b are integers.

.....

(Total for Question 13 is 2 marks)

14 Cone **A** and cone **B** are mathematically similar.
The ratio of the volume of cone **A** to the volume of cone **B** is 27 : 8

The surface area of cone **A** is 297 cm^2

Show that the surface area of cone **B** is 132 cm^2

Volume ratio A : B = 27 : 8

Length ratio = 3 : 2 (cube roots)

1 mark
Cube-root the volume ratio

Area ratio = $3^2 : 2^2 = 9 : 4$

1 mark
Square the length ratio

	A	B
×33	9	4
	297	132

×33

Area of B = 297 ÷ 9 × 4 = 132 cm^2

1 mark
Required area shown from the correct ratio

(Total for Question 14 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA